

भारतीय सूचना प्रौद्योगिकी संस्थान गुवाहाटी

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY GUWAHATI

CS 360: Machine Learning Lab
Practice Assignment 8

Instructions: This is only for practice. Complete it by 12:00 PM today. Your completion will be reviewed by the Teaching Assistants.

1. Consider the **Heart Disease dataset** to classify patients having heart disease or not based on the given predictors.
   Write a program to do the followings:
   (a) Read the dataset.
   (b) Perform feature scaling / normalization, if necessary.
   (c) Implement K-NN classifier algorithm using $K = 3, 5, 7$ to classify the patients.
   (d) Evaluate the designed classifier using 5-fold cross-validation. Report the results in terms accuracy, class-wise accuracy, precision and recall for all three $k-$values, separately.
   (e) Re-implement 1(c) and 1(d) using weighted K-NN classifier using the inverse-distance weighting formula:
   $w_i = 1/(d_i + \epsilon)$; where d_i is the distance of the i-th neighbor and ϵ is a small positive constant, like 0.01.
   (f) Compare the predictions with those of standard K-NN, and report your observations.

2. Write a program to perform classification using Iris dataset using sigmoid neurons (under one vs all framework) considering 70:10:30 from train, validation, and test splits.
   (a) Report the results of hyperparameter tuning considering validation split.
   (b) Report the class-wise accuracy, precision, recall for training and testing samples.

3. Write a program to perform classification using Iris dataset using single-layer perceptron (SLP) (in single architecture) using sigmoid activation function. Finally report class-wise precision, recall, accuracy and overall accuracy considering 5-fold cross validation.

4. Write a program to perform classification on XOR gate using Multi-layer Perceptron (MLP) algorithm with error Back-propagation **without using in-built Python packages**.
   Consider the following :
   - Activation function: Sigmoid
   - Number of hidden layer: 1
   - Number of neurons in hidden layer: 2
   - w= random initialization.
   - learning rate, α= 0.1

- Number of epochs: 2
- train: validation: test split : 70:10:20

(a) Report the class-wise accuracy, precision, recall for training and testing samples, after 2 epochs.

5. Write a program to perform classification using Pima Indian Diabetes dataset using using Multi-layer Perceptron (MLP) algorithm with error Back-propagation (**In-built python packages can be used**).
   Consider the same parameter and hyperparameter settings as Q4, except number of epochs to be 50 and 100.
   Finally report class-wise precision, recall, accuracy and overall accuracy considering 5-fold cross validation, considering both cases of epochs.